

1 Bounded reduction

The purpose of this section is to remove the boundedness hypothesis from the sharp Saint–Venant stability inequality. We assume given the following *bounded* estimate, whose proof occupies the earlier part of the manuscript and which we treat here as a black box.

Theorem 1.1 (Bounded sharp Saint–Venant). *For every $n \geq 2$ and every $R \geq 2$ there is a constant $c_{\text{SV}}(n, R) > 0$ such that*

$$E(\Omega) - E(B_1) \geq c_{\text{SV}}(n, R) \mathcal{A}(\Omega)^2 \quad (1)$$

for every open set $\Omega \subset B_R$ with $|\Omega| = \omega_n$.

Our goal is the global statement.

Theorem 1.2 (Global sharp Saint–Venant stability). *There is a dimensional constant $c_{\text{SV}}^{\text{glob}}(n) > 0$ such that*

$$E(\Omega) - E(B_1) \geq c_{\text{SV}}^{\text{glob}}(n) \mathcal{A}(\Omega)^2 \quad (2)$$

for every open set $\Omega \subset \mathbb{R}^n$ with $|\Omega| = \omega_n$. Equivalently, after rescaling, for every open set Ω of finite measure, with B^* the ball of the same volume,

$$T(B^*) - T(\Omega) = 2(E(\Omega) - E(B^*)) \geq 2c_{\text{SV}}^{\text{glob}}(n) \left(\frac{|\Omega|}{\omega_n}\right)^{(n+2)/n} \mathcal{A}(\Omega)^2. \quad (3)$$

The strategy is the constructive reduction of Brasco, De Philippis and Velichkov [1, §5]. From a set Ω with small Saint–Venant deficit we manufacture a *bounded* surrogate $\tilde{\Omega}$, of the same volume, of dimensionally bounded diameter, with comparable deficit and with controlled loss of asymmetry; applying Theorem 1.1 to $\tilde{\Omega}$ at a fixed dimensional radius and transferring the estimate back to Ω closes the small-deficit regime. The complementary large-deficit regime is trivial because $\mathcal{A} < 2$. The whole construction is explicit: no compactness, no minimising sequence and no contradiction argument enter.

The two technical ingredients are an L^∞ tail estimate for the torsion function (Lemma 1.5, proved by a De Giorgi iteration) and the truncation construction itself (Lemma 1.9). Both are proved here in full rather than cited.

1.1 Standing notation and the standard tools we cite

Throughout, $n \geq 2$ and all sets are open subsets of $\mathbb{R}^n$ of finite Lebesgue measure. We write $u_\Omega \in H_0^1(\Omega)$ for the torsion function, i.e. the unique weak solution of $-\Delta u_\Omega = 1$ in Ω , $u_\Omega = 0$ on $\partial\Omega$, equivalently the minimiser of

$$J_\Omega(v) = \frac{1}{2} \int_\Omega |\nabla v|^2 dx - \int_\Omega v dx, \quad v \in H_0^1(\Omega), \quad (4)$$

and $E(\Omega) = J_\Omega(u_\Omega) = \min_{H_0^1(\Omega)} J_\Omega$. Testing the equation with u_Ω gives the standard identities

$$\int_\Omega |\nabla u_\Omega|^2 dx = \int_\Omega u_\Omega dx = T(\Omega), \quad E(\Omega) = -\frac{1}{2} \int_\Omega u_\Omega dx = -\frac{1}{2} T(\Omega). \quad (5)$$

We use $\omega_n = |B_1|$, $T_n = T(B_1) = \omega_n/(n(n+2))$, the Fraenkel asymmetry $\mathcal{A}(\Omega) = \inf_{x_0} |\Omega \triangle (B_1 + x_0)|/\omega_n$ (for $|\Omega| = \omega_n$ this lies in $[0, 2)$), and the scale-invariant deficit

$$D(\Omega) = E(\Omega) |\Omega|^{-(n+2)/n} - E(B_1) |B_1|^{-(n+2)/n}, \quad (6)$$

so that $D(\Omega) = \omega_n^{-(n+2)/n} (E(\Omega) - E(B_1)) \geq 0$ when $|\Omega| = \omega_n$. The non-negativity of D is the *Saint-Venant inequality* (equivalently Talenti's comparison theorem): among sets of given volume the ball minimises E , i.e.

$$E(\Omega) |\Omega|^{-(n+2)/n} \geq E(B) |B|^{-(n+2)/n} \quad \text{for every ball } B. \quad (7)$$

We treat (7) as a known classical result.

The external analytic facts we invoke without proof are:

- the *Sobolev inequality* $\|w\|_{L^{2^*}(\mathbb{R}^n)} \leq S_n \|\nabla w\|_{L^2(\mathbb{R}^n)}$ for $w \in H^1(\mathbb{R}^n)$ when $n \geq 3$ (with $2^* = 2n/(n-2)$), and its $n = 2$ surrogate explained in Remark 1.6;
- the *Saint-Venant/Talenti inequality* (7), used in the truncation argument and in the global L^∞ bound (8) below;
- Theorem 1.1, the bounded stability input;
- the *suboptimal* torsional stability bound $\mathcal{A}(\Omega)^4 \leq C(n)D(\Omega)$ (Lemma 1.7), used only to seed the truncation iteration. This is the single nontrivial result we cite rather than prove; see Remark 1.8.

Everything else — in particular the fast-convergence iteration lemma (Lemma 1.3), the De Giorgi L^∞ tail estimate (Lemma 1.5), the energy-comparison inequality, the truncation iteration (Lemma 1.9) and the gluing — is proved from scratch.

We will also need the elementary pointwise sup-bound on the torsion function. By Talenti's symmetrisation comparison the torsion function of Ω is dominated, after Schwarz symmetrisation, by that of the ball of the same volume; in particular

$$\|u_\Omega\|_{L^\infty(\Omega)} \leq \|u_{B^*}\|_{L^\infty(B^*)} = \frac{(|\Omega|/\omega_n)^{2/n}}{2n} =: \Lambda_n |\Omega|^{2/n}, \quad \Lambda_n := \frac{\omega_n^{-2/n}}{2n}, \quad (8)$$

where B^* is the ball with $|B^*| = |\Omega|$ and we used $u_{B^*}(x) = (r^2 - |x|^2)/(2n)$. For $|\Omega| = \omega_n$ this reads $\|u_\Omega\|_{L^\infty(\Omega)} \leq 1/(2n)$. We record (8) as a consequence of the cited Talenti comparison.

1.2 The fast-convergence iteration lemma

The De Giorgi scheme reduces an L^∞ bound to the vanishing of a nonlinear recursion. We isolate the underlying numerical lemma and prove it, so that nothing is cited at this step. The mechanism is that a super-quadratic recursion with a small enough starting value beats any fixed geometric growth factor b^k .

Lemma 1.3 (Fast geometric convergence). *Let $(a_k)_{k \geq 0}$ be non-negative reals with*

$$a_{k+1} \leq C b^k a_k^{1+\beta}, \quad k \geq 0, \quad (9)$$

where $C > 0$, $b > 1$, $\beta > 0$. If

$$a_0 \leq C^{-1/\beta} b^{-1/\beta^2}, \quad (10)$$

then

$$a_k \leq b^{-k/\beta} a_0 \quad \text{for all } k \geq 0, \quad (11)$$

so $a_k \rightarrow 0$.

Proof. Put $r := b^{-1/\beta} \in (0, 1)$ and $\theta := C a_0^\beta$. The smallness hypothesis (10) is precisely $\theta \leq r$. We prove the target $a_k \leq r^k a_0$ by induction; this is (11), and $r < 1$ then gives $a_k \rightarrow 0$.

Base case $k = 0$: equality. *Inductive step.* Assume $a_k \leq r^k a_0$. Using (9),

$$a_{k+1} \leq C b^k a_k^{1+\beta} = C b^k a_k a_k^\beta \leq C b^k a_k (r^k a_0)^\beta = (C a_0^\beta) (b r^\beta)^k a_k = \theta (b r^\beta)^k a_k.$$

By the definition of r we have $r^\beta = b^{-1}$, hence $br^\beta = 1$ and the geometric factor collapses:

$$a_{k+1} \leq \theta a_k \leq r a_k \leq r \cdot r^k a_0 = r^{k+1} a_0,$$

using $\theta \leq r$ and the inductive hypothesis. This closes the induction. $\square$

Remark 1.4. The identity $br^\beta = 1$ with $r = b^{-1/\beta}$ is the algebraic mechanism behind the convergence: the super-quadratic gain $a_k^{1+\beta}$ converts, once a_0 is small, the growing factor b^k into a contraction.

1.3 The De Giorgi L^∞ tail estimate

Lemma 1.5 (L^∞ tail estimate). *There is a dimensional constant $C_b = C_b(n) > 0$ such that for every open Ω with $|\Omega| = \omega_n$ and every $R \geq 1$,*

$$\|u_\Omega\|_{L^\infty(\Omega \setminus B_{R+1})} \leq C_b |\Omega \setminus B_R|^{1/n}. \quad (12)$$

Proof. If $|\Omega \setminus B_R| = 0$ then $u_\Omega = 0$ a.e. on $\Omega \setminus B_R \supset \Omega \setminus B_{R+1}$ and (12) is trivial, so assume $|\Omega \setminus B_R| > 0$. We carry out a De Giorgi iteration on the annular region $\Omega \setminus B_R$.

Set-up: shrinking radii, cut-offs, and rising levels. For $k \in \mathbb{N}_0$ put

$$R_k := R + 1 - 2^{-k},$$

so that $R_0 = R$, the radii increase to $R_\infty = R + 1$, and $R_k \geq R \geq 1$. Let $\varphi_k \in C^{0,1}(\mathbb{R}^n)$ be the radial cut-off

$$\varphi_k(x) := \phi_k(|x|), \quad \phi_k = 0 \text{ on } [0, R_{k-1}], \quad \phi_k = 1 \text{ on } [R_k, \infty), \quad \phi_k \text{ affine on } [R_{k-1}, R_k],$$

for $k \geq 1$, and $\varphi_0 \equiv 1$ on $\{|x| \geq R_0\}$, extended by 0 inside (we only use φ_k for $k \geq 1$ in the recursion). Then $0 \leq \varphi_k \leq 1$, $\varphi_k \equiv 0$ on $B_{R_{k-1}}$, $\varphi_k \equiv 1$ on $\mathbb{R}^n \setminus B_{R_k}$, and

$$|\nabla \varphi_k| \leq \frac{1}{R_k - R_{k-1}} = \frac{1}{2^{-k} - 2^{-(k+1)}} = 2^{k+1}.$$

Choose rising levels

$$s_k := M |\Omega \setminus B_R|^{1/n} (1 - 2^{-k}), \quad k \geq 0,$$

where $M = M(n) \geq 1$ is a free constant fixed at the end. Thus $s_0 = 0$, $s_k \uparrow s_\infty := M |\Omega \setminus B_R|^{1/n}$, and

$$s_{k+1} - s_k = M |\Omega \setminus B_R|^{1/n} 2^{-(k+1)}. \quad (13)$$

Caccioppoli inequality from the truncated test function. Fix $k \geq 1$. The function $(u_\Omega - s_k)_+ \in H_0^1(\Omega)$ (it vanishes where $u_\Omega \leq s_k$, in particular near $\partial\Omega$ since $u_\Omega \geq 0$ and $s_k \geq 0$), and φ_k is bounded Lipschitz, so

$$v_k := \varphi_k^2 (u_\Omega - s_k)_+ \in H_0^1(\Omega)$$

is an admissible test function. Insert it into the weak formulation $\int_\Omega \nabla u_\Omega \cdot \nabla v \, dx = \int_\Omega v \, dx$. Using $\nabla v_k = \varphi_k^2 \nabla (u_\Omega - s_k)_+ + 2\varphi_k (u_\Omega - s_k)_+ \nabla \varphi_k$ and $\nabla u_\Omega \cdot \nabla (u_\Omega - s_k)_+ = |\nabla (u_\Omega - s_k)_+|^2$ (the gradient of u_Ω equals that of $(u_\Omega - s_k)_+$ on $\{u_\Omega > s_k\}$ and both factors vanish elsewhere), we get

$$\int \varphi_k^2 |\nabla (u_\Omega - s_k)_+|^2 \, dx + 2 \int \varphi_k (u_\Omega - s_k)_+ \nabla \varphi_k \cdot \nabla (u_\Omega - s_k)_+ \, dx = \int \varphi_k^2 (u_\Omega - s_k)_+ \, dx.$$

The middle term is bounded by Young's inequality, $2 |\varphi_k (u_\Omega - s_k)_+ \nabla \varphi_k \cdot \nabla (u_\Omega - s_k)_+| \leq \frac{1}{2} \varphi_k^2 |\nabla (u_\Omega - s_k)_+|^2 + 2 |\nabla \varphi_k|^2 (u_\Omega - s_k)_+^2$, and absorbing the first piece on the left yields the Caccioppoli estimate

$$\int |\nabla (\varphi_k (u_\Omega - s_k)_+)|^2 \, dx \leq C_1 \int |\nabla \varphi_k|^2 (u_\Omega - s_k)_+^2 \, dx + C_1 \int \varphi_k^2 (u_\Omega - s_k)_+ \, dx, \quad (14)$$

with an absolute constant C_1 (one checks $C_1 = 4$ works after expanding $|\nabla(\varphi_k w)|^2 \leq 2\varphi_k^2 |\nabla w|^2 + 2|\nabla\varphi_k|^2 w^2$ and combining with the absorbed bound; the precise value is immaterial). Here we wrote $w = (u_\Omega - s_k)_+$.

Sobolev inequality. Set $w_k := \varphi_k(u_\Omega - s_k)_+ \in H_0^1(\mathbb{R}^n)$ and

$$A_k := \{\varphi_k(u_\Omega - s_k)_+ > 0\} = \{u_\Omega > s_k\} \cap \{\varphi_k > 0\} \subset \{u_\Omega > s_k\} \cap (\mathbb{R}^n \setminus B_{R_{k-1}}).$$

For $n \geq 3$ the Sobolev inequality and Hölder's inequality on the support A_k give

$$\int w_k^2 dx \leq \left(\int w_k^{2^*} dx \right)^{2/2^*} |A_k|^{1-2/2^*} \leq S_n^2 |A_k|^{2/n} \int |\nabla w_k|^2 dx, \quad (15)$$

using $1 - 2/2^* = 2/n$. (The $n = 2$ case is handled in Remark 1.6; the resulting estimate has the same shape with $2/n$ replaced by any fixed exponent in $(0, 1)$, which only changes the dimensional constants.)

Combining (15) with (14), $|\nabla\varphi_k| \leq 2^{k+1}$ (so $|\nabla\varphi_k|^2 \leq 4^{k+1}$) and $0 \leq \varphi_k \leq 1$, we obtain

$$\int w_k^2 dx \leq S_n^2 C_1 |A_k|^{2/n} \left(4^{k+1} \int_{A_k} (u_\Omega - s_k)_+^2 dx + \int_{A_k} (u_\Omega - s_k)_+ dx \right), \quad (16)$$

where for the second term we used $\int \varphi_k^2 (u_\Omega - s_k)_+ \leq \int_{A_k} (u_\Omega - s_k)_+$. By the global sup-bound (8) at $|\Omega| = \omega_n$ we have $0 \leq (u_\Omega - s_k)_+ \leq u_\Omega \leq 1/(2n) \leq 1$ on Ω , hence both $(u_\Omega - s_k)_+^2 \leq 1$ and $(u_\Omega - s_k)_+ \leq 1$ there, so each integral over A_k is at most $|A_k|$. Therefore

$$\int w_k^2 dx \leq S_n^2 C_1 (4^{k+1} + 1) |A_k|^{2/n} |A_k| \leq C_2 4^k |A_k|^{1+2/n}, \quad (17)$$

with $C_2 = C_2(n) := 8 S_n^2 C_1$ (using $4^{k+1} + 1 \leq 8 \cdot 4^k$).

From energy to a measure recursion. Restrict the left-hand side of (17) to the smaller set where the *next* level is exceeded. On $\{u_\Omega > s_{k+1}\} \cap (\mathbb{R}^n \setminus B_{R_k})$ we have $\varphi_k = 1$ (since this set lies outside B_{R_k}) and $u_\Omega - s_k > s_{k+1} - s_k$, hence $w_k^2 = (u_\Omega - s_k)_+^2 \geq (s_{k+1} - s_k)^2$ there. Therefore

$$(s_{k+1} - s_k)^2 |\{u_\Omega > s_{k+1}\} \cap (\mathbb{R}^n \setminus B_{R_k})| \leq \int_{\mathbb{R}^n} w_k^2 dx. \quad (18)$$

Introduce the (relative) super-level measures

$$\mu_k := |\{u_\Omega > s_k\} \cap (\mathbb{R}^n \setminus B_{R_{k-1}})|, \quad k \geq 1,$$

which decrease in k (both s_k and R_{k-1} increase). Since $A_k \subset \{u_\Omega > s_k\} \cap (\mathbb{R}^n \setminus B_{R_{k-1}})$ we have $|A_k| \leq \mu_k$, and the left-hand set in (18) is exactly $\{u_\Omega > s_{k+1}\} \cap (\mathbb{R}^n \setminus B_{R_{(k+1)-1}})$, of measure μ_{k+1} . Combining (18), (17) and $|A_k| \leq \mu_k$,

$$(s_{k+1} - s_k)^2 \mu_{k+1} \leq C_2 4^k \mu_k^{1+2/n}.$$

Insert the gap (13), $(s_{k+1} - s_k)^2 = M^2 |\Omega \setminus B_R|^{2/n} 4^{-(k+1)}$:

$$\mu_{k+1} \leq \frac{C_2 4^k}{M^2 |\Omega \setminus B_R|^{2/n} 4^{-(k+1)}} \mu_k^{1+2/n} = \frac{4 C_2}{M^2 |\Omega \setminus B_R|^{2/n}} 16^k \mu_k^{1+2/n}. \quad (19)$$

Normalisation and the iteration lemma. Set $a_k := \mu_k / |\Omega \setminus B_R|$. Since $\mu_k \leq |\mathbb{R}^n \setminus B_{R_{k-1}} \cap \{u_\Omega > s_k\}| \leq |\Omega \setminus B_R|$ (because $R_{k-1} \geq R_0 = R$ and $\{u_\Omega > s_k\} \subset \Omega$), we have $0 \leq a_k \leq 1$; in particular $a_1 \leq 1$. Dividing (19) by $|\Omega \setminus B_R|$ and using $\mu_k^{1+2/n} = |\Omega \setminus B_R|^{1+2/n} a_k^{1+2/n}$,

$$a_{k+1} \leq \frac{4 C_2}{M^2 |\Omega \setminus B_R|^{2/n}} 16^k \frac{|\Omega \setminus B_R|^{1+2/n}}{|\Omega \setminus B_R|} a_k^{1+2/n} = \frac{4 C_2}{M^2} 16^k a_k^{1+2/n}.$$

valid for all $k \geq 1$. To put this in the exact form (9) (indexed from 0) we shift: set $\tilde{a}_j := a_{j+1}$ for $j \geq 0$. Then for every $j \geq 0$,

$$\tilde{a}_{j+1} = a_{j+2} \leq \frac{4C_2}{M^2} 16^{j+1} a_{j+1}^{1+2/n} = \underbrace{\frac{64C_2}{M^2}}_{=:C} 16^j \tilde{a}_j^{1+2/n},$$

which is (9) with

$$C := \frac{64C_2}{M^2}, \quad b := 16, \quad \beta := \frac{2}{n}.$$

Choose $M = M(n)$ so large that the smallness condition (10) holds at the starting value $\tilde{a}_0 = a_1 \leq 1$. Since $a_1 \leq 1$ it suffices to require

$$1 \leq C^{-1/\beta} b^{-1/\beta^2} = \left(\frac{64C_2}{M^2}\right)^{-n/2} 16^{-n^2/4}, \quad \text{i.e.} \quad M^2 \geq 64C_2 16^{n^2/4 \cdot (2/n)} = 64C_2 16^{n/2},$$

which holds for $M := 8(C_2)^{1/2} 4^{n/2}$ (a dimensional constant, since $C_2 = C_2(n)$). With this M , Lemma 1.3 gives $\tilde{a}_j \rightarrow 0$, i.e. $a_k \rightarrow 0$ and hence $\mu_k \rightarrow 0$.

Conclusion. Letting $k \rightarrow \infty$ in $\mu_k \rightarrow 0$ and using $s_k \uparrow s_\infty = M |\Omega \setminus B_R|^{1/n}$ and $R_{k-1} \uparrow R+1$, monotone convergence yields

$$|\{u_\Omega > s_\infty\} \cap (\mathbb{R}^n \setminus B_{R+1})| = \lim_{k \rightarrow \infty} \mu_k = 0.$$

Hence $u_\Omega \leq s_\infty = M |\Omega \setminus B_R|^{1/n}$ a.e. on $\Omega \setminus B_{R+1}$. Since $u_\Omega \geq 0$, this is (12) with $C_b := M = 8(C_2)^{1/2} 4^{n/2}$. $\square$

Remark 1.6 (The case $n = 2$). For $n = 2$ the embedding $H_0^1 \hookrightarrow L^{2^*}$ degenerates, but the Gagliardo–Nirenberg–Sobolev inequality gives, for any fixed $p \in (2, \infty)$, $\|w\|_{L^p(\mathbb{R}^2)} \leq S_{2,p} \|\nabla w\|_{L^2(\mathbb{R}^2)}$ for $w \in H_0^1(\mathbb{R}^2)$ (e.g. via the $n = 2$ Ladyzhenskaya/Sobolev embedding, valid on bounded support). Repeating the Hölder step with exponent p replaces the exponent $2/n = 1$ by $1 - 2/p \in (0, 1)$ in (15)–(19); the recursion (19) becomes $\mu_{k+1} \leq C 16^k \mu_k^{1+(1-2/p)}$ and Lemma 1.3 applies verbatim with $\beta = 1 - 2/p > 0$. The final exponent on $|\Omega \setminus B_R|$ in (12) stays $1/n = 1/2$ because it is fixed by the level scale $s_\infty = M |\Omega \setminus B_R|^{1/n}$, which we are free to choose; only the dimensional constant C_b changes. Thus (12) holds for all $n \geq 2$.

1.4 A suboptimal stability seed

The truncation iteration needs one quantitative input that is genuinely *not* produced by the De Giorgi machinery: a (possibly far from sharp) quantitative control of the asymmetry by the deficit, used only to guarantee that the tail mass is already small at the first radius and to bound the number of truncation steps. We state it with the exponent 4; this is the one ingredient of the present section that we cite rather than prove, and we record this honestly.

Lemma 1.7 (Suboptimal stability, cited). *There is a dimensional constant $C_\dagger = C_\dagger(n) > 0$ such that every open $\Omega \subset \mathbb{R}^n$ of finite measure satisfies*

$$D(\Omega) \geq \frac{1}{C_\dagger} \mathcal{A}(\Omega)^4. \quad (20)$$

Equivalently, $\mathcal{A}(\Omega) \leq (C_\dagger D(\Omega))^{1/4}$.

Remark 1.8 (Status of Lemma 1.7). Inequality (20) is the torsional stability estimate of Fusco–Maggi–Pratelli [2], obtained from the Pólya–Szegő inequality and the quantitative isoperimetric inequality; via $E(\Omega) = -1/(2\lambda_{2,1}(\Omega))$ it is exactly [1, Lemma 5.2]. Its proof is rearrangement-based (no compactness, no contradiction), but it lies outside the De Giorgi–truncation circle that is the subject of this section, so we import it. Only a positive power of D is used below; the exponent 4 is not essential, any fixed exponent would do.

1.5 The truncation construction

We now build the bounded surrogate. The estimate driving the construction is an *energy-comparison inequality*: cutting Ω off at radius $\sim k$ costs only a higher-order amount of energy, controlled by the tail mass through Lemma 1.5.

Lemma 1.9 (Bounded truncation). *There exist dimensional constants $C_\sharp = C_\sharp(n) > 0$, $\delta = \delta(n) \in (0, 1]$ and $d = d(n) \geq 1$ such that for every open Ω with $|\Omega| = \omega_n$ and $D(\Omega) \leq \delta$ there is an open set $\tilde{\Omega}$ with*

$$|\tilde{\Omega}| = \omega_n, \quad (21)$$

$$\text{diam}(\tilde{\Omega}) \leq d, \quad (22)$$

$$D(\tilde{\Omega}) \leq C_\sharp D(\Omega), \quad (23)$$

$$\mathcal{A}(\Omega) \leq \mathcal{A}(\tilde{\Omega}) + C_\sharp D(\Omega). \quad (24)$$

Proof. Normalise so that the ball realising the asymmetry of Ω is B_1 (translate Ω ; both $\mathcal{A}$ and D are translation invariant), i.e. $\mathcal{A}(\Omega) = |\Omega \triangle B_1|/\omega_n$. We will repeatedly use the tail-mass notation

$$b_k := \frac{|\Omega \setminus B_k|}{\omega_n}, \quad k \geq 1, \quad (25)$$

which is non-increasing in k and satisfies $0 \leq b_k \leq 1$.

Step 0: the tail mass is small. Since B_1 realises the asymmetry, $|\Omega \setminus B_1| \leq |\Omega \triangle B_1| = \omega_n \mathcal{A}(\Omega)$. Combined with the suboptimal seed Lemma 1.7,

$$b_1 \leq \mathcal{A}(\Omega) \leq (C_\dagger D(\Omega))^{1/4}. \quad (26)$$

In particular, since $b_k \leq b_1$ for all $k \geq 1$, every tail mass is $\leq (C_\dagger D(\Omega))^{1/4}$, which can be made smaller than any prescribed dimensional threshold by shrinking $\delta(n)$ (recall $D(\Omega) \leq \delta(n)$). This is the only use of Lemma 1.7 in the construction.

Step 1: the energy-comparison inequality. For $k \geq 1$ let φ_k be the Lipschitz cut-off

$$\varphi_k(x) := \min\{1, (k+2 - |x|)_+\},$$

so $\varphi_k \equiv 1$ on B_{k+1} , $\varphi_k \equiv 0$ outside B_{k+2} , $0 \leq \varphi_k \leq 1$ and $|\nabla \varphi_k| \leq 1$ everywhere (it is 1 only on the annulus $B_{k+2} \setminus B_{k+1}$). Then $u_k := \varphi_k u_\Omega \in H_0^1(\Omega \cap B_{k+2})$ is admissible for the variational problem defining $E(\Omega \cap B_{k+2})$, so $E(\Omega \cap B_{k+2}) \leq J_{\Omega \cap B_{k+2}}(u_k) = \frac{1}{2} \int |\nabla u_k|^2 - \int u_k$. We use the pointwise algebraic identity

$$|\nabla(\varphi_k u_\Omega)|^2 = \nabla u_\Omega \cdot \nabla(u_\Omega \varphi_k^2) + |\nabla \varphi_k|^2 u_\Omega^2,$$

which is verified by expanding both sides ($\nabla(u_\Omega \varphi_k^2) = \varphi_k^2 \nabla u_\Omega + 2\varphi_k u_\Omega \nabla \varphi_k$, so the right-hand side equals $\varphi_k^2 |\nabla u_\Omega|^2 + 2\varphi_k u_\Omega \nabla u_\Omega \cdot \nabla \varphi_k + |\nabla \varphi_k|^2 u_\Omega^2 = |\varphi_k \nabla u_\Omega + u_\Omega \nabla \varphi_k|^2$). Integrating and applying the weak equation $\int \nabla u_\Omega \cdot \nabla(u_\Omega \varphi_k^2) = \int u_\Omega \varphi_k^2$ with the admissible test function $u_\Omega \varphi_k^2 \in H_0^1(\Omega)$, we get

$$J_{\Omega \cap B_{k+2}}(u_k) = \frac{1}{2} \int u_\Omega \varphi_k^2 + \frac{1}{2} \int |\nabla \varphi_k|^2 u_\Omega^2 - \int \varphi_k u_\Omega.$$

Now use $\frac{1}{2} \varphi_k^2 - \varphi_k = -\frac{1}{2}(1 - (1 - \varphi_k)^2) = -\frac{1}{2} + \frac{1}{2}(1 - \varphi_k)^2$ to write $\frac{1}{2} \int u_\Omega \varphi_k^2 - \int \varphi_k u_\Omega = -\frac{1}{2} \int u_\Omega + \frac{1}{2} \int (1 - \varphi_k)^2 u_\Omega = E(\Omega) + \frac{1}{2} \int (1 - \varphi_k)^2 u_\Omega$, where we used $E(\Omega) = -\frac{1}{2} \int u_\Omega$ from (5). Therefore

$$E(\Omega \cap B_{k+2}) \leq J_{\Omega \cap B_{k+2}}(u_k) = E(\Omega) + \frac{1}{2} \int (1 - \varphi_k)^2 u_\Omega + \frac{1}{2} \int |\nabla \varphi_k|^2 u_\Omega^2. \quad (27)$$

Both correction integrals are supported in $\mathbb{R}^n \setminus B_{k+1}$: indeed $1 - \varphi_k = 0$ on B_{k+1} and $\nabla \varphi_k = 0$ on $B_{k+1} \cup (\mathbb{R}^n \setminus B_{k+2})$. Hence, with $0 \leq 1 - \varphi_k \leq 1$ and $|\nabla \varphi_k| \leq 1$,

$$E(\Omega \cap B_{k+2}) - E(\Omega) \leq \frac{1}{2} |\Omega \setminus B_{k+1}| \left(\sup_{\Omega \setminus B_{k+1}} u_\Omega + \sup_{\Omega \setminus B_{k+1}} u_\Omega^2 \right). \quad (28)$$

By Lemma 1.5 applied with $R = k \geq 1$ (so $R + 1 = k + 1$ and the tail region is $\Omega \setminus B_{k+1}$), $\sup_{\Omega \setminus B_{k+1}} u_\Omega \leq C_b |\Omega \setminus B_k|^{1/n} = C_b \omega_n^{1/n} b_k^{1/n}$, and likewise the square is bounded by $C_b^2 \omega_n^{2/n} b_k^{2/n}$. Also $|\Omega \setminus B_{k+1}| = \omega_n b_{k+1} \leq \omega_n b_k$. Plugging in, and using $b_k \leq 1$ so that $b_k^{2/n} \leq b_k^{1/n}$,

$$E(\Omega \cap B_{k+2}) \leq E(\Omega) + C b_{k+1} (b_k^{1/n} + b_k^{2/n}) \leq E(\Omega) + C b_k^{1+1/n}, \quad (29)$$

for a dimensional constant $C = C(n)$ (we absorbed $\frac{1}{2} \omega_n (C_b \omega_n^{1/n} + C_b^2 \omega_n^{2/n})$ and $b_{k+1} \leq b_k$ into C). This is the energy-comparison inequality.

Step 2: turning energy comparison into a tail recursion. By the Saint-Venant inequality (7) applied to the set $\Omega \cap B_{k+2}$, whose volume is $|\Omega \cap B_{k+2}| = \omega_n (1 - b_{k+2})$,

$$E(B_1) |B_1|^{-(n+2)/n} \leq E(\Omega \cap B_{k+2}) |\Omega \cap B_{k+2}|^{-(n+2)/n},$$

that is

$$E(B_1) \omega_n^{-(n+2)/n} (\omega_n (1 - b_{k+2}))^{(n+2)/n} \leq E(\Omega \cap B_{k+2}),$$

i.e.

$$E(B_1) (1 - b_{k+2})^{(n+2)/n} \leq E(\Omega \cap B_{k+2}).$$

Combine with (29) and the definition $E(\Omega) = E(B_1) + \omega_n^{(n+2)/n} D(\Omega)$ (which is (6) at $|\Omega| = \omega_n$):

$$E(B_1) (1 - b_{k+2})^{(n+2)/n} \leq E(\Omega) + C b_k^{1+1/n} = E(B_1) + \omega_n^{(n+2)/n} D(\Omega) + C b_k^{1+1/n}.$$

Rearranging, and recalling $E(B_1) < 0$, divide by $-E(B_1) > 0$:

$$1 - (1 - b_{k+2})^{(n+2)/n} \leq \frac{\omega_n^{(n+2)/n} D(\Omega) + C b_k^{1+1/n}}{-E(B_1)} =: \widehat{C}_1 (D(\Omega) + b_k^{1+1/n}),$$

with $\widehat{C}_1 = \widehat{C}_1(n) > 0$ (here $\widehat{C}_1 = \max\{\omega_n^{(n+2)/n}, C\} / (-E(B_1))$). To extract b_{k+2} we use the elementary lower bound

$$1 - (1 - t)^p \geq t \quad \text{for all } t \in [0, 1], \quad p \geq 1, \quad (30)$$

which holds because $r \mapsto (1 - t)^r$ is non-increasing on $[0, \infty)$, so $(1 - t)^p \leq (1 - t)^1 = 1 - t$ for $p = (n + 2)/n \geq 1$. Applying (30) with $t = b_{k+2} \in [0, 1]$ and $p = (n + 2)/n$,

$$b_{k+2} \leq 1 - (1 - b_{k+2})^{(n+2)/n} \leq \widehat{C} (D(\Omega) + b_k^{1+1/n}), \quad \widehat{C} := \widehat{C}_1. \quad (31)$$

Step 3: the geometric iteration to a bounded stopping index. Define the stopping index

$$K := \max\{k \geq 1 : b_k \geq 2\widehat{C} D(\Omega)\}, \quad (32)$$

with the convention $K := 0$ if $b_1 < 2\widehat{C} D(\Omega)$ (then the tail is already controlled at the first radius and Step 5 applies with $K = 0$). The set in (32) is finite because $b_k \rightarrow 0$ as $k \rightarrow \infty$ (the tail mass of a finite-measure set vanishes), so K is well defined. We prove

$$K \leq K(n) \quad (33)$$

for a dimensional $K(n)$, provided $\delta(n)$ is small enough. Assume $K \geq 3$ (otherwise (33) is trivial).

Self-improving contraction. For every k with $k+2 \leq K$ we have, by definition of K , that $b_{k+2} \geq 2\widehat{C}D(\Omega)$, i.e. $\widehat{C}D(\Omega) \leq \frac{1}{2}b_{k+2}$. Feeding this into (31),

$$b_{k+2} \leq \widehat{C}D(\Omega) + \widehat{C}b_k^{1+1/n} \leq \frac{1}{2}b_{k+2} + \widehat{C}b_k^{1+1/n},$$

and absorbing $\frac{1}{2}b_{k+2}$ on the left,

$$b_{k+2} \leq 2\widehat{C}b_k^{1+1/n} \quad \text{whenever } k+2 \leq K. \quad (34)$$

Smallness from the seed. By Step 0, $b_1 \leq (C_{\dagger}D(\Omega))^{1/4} \leq (C_{\dagger}\delta(n))^{1/4}$. Choose $\delta(n)$ small enough that

$$b_1 \leq (C_{\dagger}\delta(n))^{1/4} \leq (2\widehat{C})^{-2n} =: \eta_n \in (0, 1). \quad (35)$$

Then $2\widehat{C}b_k^{1/(2n)} \leq 2\widehat{C}\eta_n^{1/(2n)} = 1$ for all $k \geq 1$ (as $b_k \leq b_1 \leq \eta_n$), so (34) self-improves to

$$b_{k+2} = (2\widehat{C}b_k^{1/(2n)})b_k^{1+1/(2n)} \leq b_k^{1+1/(2n)} \quad \text{whenever } k+2 \leq K. \quad (36)$$

Iterating along odd indices. Apply (36) repeatedly, starting from $k=1$, as long as the index stays $\leq K$. Writing the odd indices $1, 3, 5, \dots$ and setting $\gamma := 1 + \frac{1}{2n} > 1$, induction gives, for every $j \geq 0$ with $2j+1 \leq K$,

$$b_{2j+1} \leq b_1^{\gamma^j}. \quad (37)$$

Indeed (37) holds at $j=0$, and if it holds at j with $2(j+1)+1 = 2j+3 \leq K$, then (36) (at $k=2j+1$, legitimate since $2j+3 \leq K$) gives $b_{2j+3} \leq b_{2j+1}^{\gamma} \leq (b_1^{\gamma^j})^{\gamma} = b_1^{\gamma^{j+1}}$.

Bounding K . Let $j_K := \lfloor (K-1)/2 \rfloor$, the largest j with $2j+1 \leq K$. By the maximality of K and (37),

$$2\widehat{C}D(\Omega) \leq b_{2j_K+1} \leq b_1^{\gamma^{j_K}} \leq (C_{\dagger}D(\Omega))^{\gamma^{j_K}/4}, \quad (38)$$

using $b_1 \leq (C_{\dagger}D(\Omega))^{1/4}$. Take logarithms (recall $D(\Omega) \leq \delta(n) < 1$, so $\log D(\Omega) < 0$); writing $L := -\log D(\Omega) > 0$, (38) reads

$$\log(2\widehat{C}) - L \leq \frac{\gamma^{j_K}}{4}(\log C_{\dagger} - L).$$

For $\delta(n)$ small, $L = -\log D(\Omega) \geq -\log \delta(n)$ is large; in particular we may take $\delta(n)$ small enough that $L \geq 2 \max\{|\log(2\widehat{C})|, |\log C_{\dagger}|\}$, whence $\log(2\widehat{C}) - L \geq -\frac{3}{2}L$ and $\log C_{\dagger} - L \leq -\frac{1}{2}L$.

The displayed inequality then forces $-\frac{3}{2}L \leq -\frac{\gamma^{j_K}}{8}L$, i.e. $\gamma^{j_K} \leq 12$. Since $\gamma = 1 + \frac{1}{2n} > 1$ this bounds $j_K \leq \log 12 / \log \gamma =: j_*(n)$, hence $K \leq 2j_K + 2 \leq 2j_*(n) + 2 =: K(n)$, a dimensional constant. This proves (33).

Tail at the stopping radius. By the maximality (32), $b_{K+1} < 2\widehat{C}D(\Omega)$; we record this for the next step.

Step 5: the truncated set and its volume. By the definition (32) of K and $K \leq K(n)$, $b_{K+1} < 2\widehat{C}D(\Omega)$, hence

$$|\Omega \setminus B_{K+1}| = \omega_n b_{K+1} < 2\widehat{C}\omega_n D(\Omega). \quad (39)$$

In fact we cut one radius further out to have room for the cut-off: by monotonicity $b_{K+3} \leq b_{K+1} < 2\widehat{C}D(\Omega)$, so

$$|\Omega \cap B_{K+3}| = \omega_n(1 - b_{K+3}) \geq \omega_n(1 - 2\widehat{C}D(\Omega)), \quad |\Omega \setminus B_{K+3}| < 2\widehat{C}\omega_n D(\Omega). \quad (40)$$

Define

$$r := \left(\frac{|\Omega \cap B_{K+3}|}{\omega_n} \right)^{1/n} = (1 - b_{K+3})^{1/n} \in [(1 - 2\widehat{C}D(\Omega))^{1/n}, 1], \quad \widetilde{\Omega} := r^{-1}(\Omega \cap B_{K+3}).$$

Then $|\tilde{\Omega}| = r^{-n} |\Omega \cap B_{K+3}| = \omega_n$, which is (21). Since $\Omega \cap B_{K+3} \subset B_{K+3}$ and $r \geq (1 - 2\hat{C}\delta(n))^{1/n} \geq \frac{1}{2}$ for $\delta(n)$ small,

$$\text{diam}(\tilde{\Omega}) = r^{-1} \text{diam}(\Omega \cap B_{K+3}) \leq r^{-1} \cdot 2(K+3) \leq 4(K(n)+3) =: d(n),$$

giving (22). We record for later the linearised bound

$$0 \leq 1 - r^n = b_{K+3} < 2\hat{C}D(\Omega), \quad 0 \leq 1 - r \leq 1 - r^n < 2\hat{C}D(\Omega), \quad (41)$$

the middle inequality because $r \leq 1$ implies $1 - r \leq 1 - r^n$.

Step 6: deficit of the surrogate. Apply the energy comparison (29) at $k = K+1$ (so the cut radius is $k+2 = K+3$):

$$E(\Omega \cap B_{K+3}) \leq E(\Omega) + C b_{K+1}^{1+1/n} \leq E(\Omega) + C (2\hat{C}D(\Omega))^{1+1/n} \leq E(\Omega) + C' D(\Omega),$$

where in the last step we used $D(\Omega) \leq \delta(n) \leq 1$ so that $D(\Omega)^{1+1/n} \leq D(\Omega)$, and absorbed constants into $C' = C'(n)$. Since E scales as $E(t\Omega') = t^{n+2}E(\Omega')$ and $\tilde{\Omega} = r^{-1}(\Omega \cap B_{K+3})$,

$$E(\tilde{\Omega}) = r^{-(n+2)} E(\Omega \cap B_{K+3}) \leq r^{-(n+2)} (E(\Omega) + C' D(\Omega)).$$

Because $|\tilde{\Omega}| = \omega_n$, $D(\tilde{\Omega}) = \omega_n^{-(n+2)/n} (E(\tilde{\Omega}) - E(B_1))$. Using $E(\Omega) = E(B_1) + \omega_n^{(n+2)/n} D(\Omega)$ and $r \leq 1$ (so $r^{-(n+2)} \geq 1$), with $E(B_1) < 0$,

$$\begin{aligned} E(\tilde{\Omega}) - E(B_1) &\leq r^{-(n+2)} (E(B_1) + \omega_n^{(n+2)/n} D(\Omega) + C' D(\Omega)) - E(B_1) \\ &= (r^{-(n+2)} - 1) E(B_1) + r^{-(n+2)} (\omega_n^{(n+2)/n} + C') D(\Omega). \end{aligned}$$

For the first term, $r^{-(n+2)} - 1 \geq 0$ and $E(B_1) < 0$ would make it negative—good for an upper bound—but we must be careful, since a negative contribution only helps. Quantitatively, by (41), $r^n \geq 1 - 2\hat{C}D(\Omega)$, so $r^{-(n+2)} = r^{-n} \cdot r^{-2} \leq (1 - 2\hat{C}D(\Omega))^{-(n+2)/n} \leq 1 + C'' D(\Omega)$ for $\delta(n)$ small (Bernoulli/Taylor bound, since $2\hat{C}D(\Omega) \leq 2\hat{C}\delta(n) \leq \frac{1}{2}$). Hence $r^{-(n+2)} - 1 \leq C'' D(\Omega)$ and $r^{-(n+2)} \leq 1 + C'' D(\Omega) \leq 2$, giving

$$E(\tilde{\Omega}) - E(B_1) \leq C'' D(\Omega) \cdot |E(B_1)| + 2(\omega_n^{(n+2)/n} + C') D(\Omega) \leq C'_2 D(\Omega),$$

with $C'_2 = C'_2(n)$. Dividing by $\omega_n^{(n+2)/n}$,

$$D(\tilde{\Omega}) = \omega_n^{-(n+2)/n} (E(\tilde{\Omega}) - E(B_1)) \leq \omega_n^{-(n+2)/n} C'_2 D(\Omega) =: C_3 D(\Omega), \quad (42)$$

which is (23) with $C_3 = C_3(n)$.

Step 7: asymmetry loss. Let $B_1(x_0)$ realise the asymmetry of $\tilde{\Omega}$, so $\mathcal{A}(\tilde{\Omega}) = |\tilde{\Omega} \triangle B_1(x_0)| / \omega_n$. We compare the asymmetry of Ω to that of $\tilde{\Omega}$ through the chain of inclusions relating Ω , $\Omega \cap B_{K+3}$ and $\tilde{\Omega} = r^{-1}(\Omega \cap B_{K+3})$. For any test ball $B_1(y)$,

$$|\Omega \triangle B_1(y)| \leq |\Omega \triangle (\Omega \cap B_{K+3})| + |(\Omega \cap B_{K+3}) \triangle B_1(y)|.$$

The first term is $|\Omega \setminus B_{K+3}| < 2\hat{C}\omega_n D(\Omega)$ by (40). For the second, scale by r : writing $E_r := \Omega \cap B_{K+3} = r\tilde{\Omega}$ and $B_1(y) = r B_{1/r}(y/r)$,

$$|(r\tilde{\Omega}) \triangle (r B_{1/r}(y/r))| = r^n |\tilde{\Omega} \triangle B_{1/r}(y/r)| \leq r^n \left(|\tilde{\Omega} \triangle B_1(y/r)| + |B_1(y/r) \triangle B_{1/r}(y/r)| \right).$$

The last term is the volume difference of concentric balls of radii 1 and $1/r \geq 1$: $|B_1 \triangle B_{1/r}| = \omega_n(r^{-n} - 1)$. Hence

$$r^n |B_1(y/r) \triangle B_{1/r}(y/r)| = r^n \omega_n(r^{-n} - 1) = \omega_n(1 - r^n) < 2\hat{C}\omega_n D(\Omega),$$

using (41). Choosing $y := rx_0$ so that $y/r = x_0$ is the optimal centre for $\tilde{\Omega}$, the middle term becomes $r^n \left| \tilde{\Omega} \triangle B_1(x_0) \right| = r^n \omega_n \mathcal{A}(\tilde{\Omega}) \leq \omega_n \mathcal{A}(\tilde{\Omega})$ (as $r \leq 1$). Collecting,

$$|\Omega \triangle B_1(rx_0)| \leq 2\hat{C}\omega_n D(\Omega) + \omega_n \mathcal{A}(\tilde{\Omega}) + 2\hat{C}\omega_n D(\Omega),$$

and dividing by ω_n and taking the infimum over translates (which only decreases the left side),

$$\mathcal{A}(\Omega) \leq \frac{|\Omega \triangle B_1(rx_0)|}{\omega_n} \leq \mathcal{A}(\tilde{\Omega}) + 4\hat{C} D(\Omega). \quad (43)$$

This is (24) with the constant $4\hat{C}$. Taking $C_{\sharp} := \max\{C_3, 4\hat{C}\} = C_{\sharp}(n)$ proves (23)–(24) simultaneously, and we already have (21)–(22). This completes the proof. $\square$

Remark 1.10 (Translation into a fixed ball). The diameter bound (22) is translation invariant, so after a translation we may assume $\tilde{\Omega} \subset B_{d(n)}$. Both $\mathcal{A}$ and D are translation invariant, so (21)–(24) are preserved.

1.6 The far branch

Lemma 1.11 (Large-deficit quadratic bound). *For every open Ω with $|\Omega| = \omega_n$ and $D(\Omega) \geq \delta(n)/4$,*

$$E(\Omega) - E(B_1) \geq \omega_n^{(n+2)/n} \frac{\delta(n)}{16} \mathcal{A}(\Omega)^2. \quad (44)$$

Proof. Since $|\Omega| = \omega_n$ gives $\mathcal{A}(\Omega) < 2$, we have $\mathcal{A}(\Omega)^2/4 < 1$. Hence

$$D(\Omega) \geq \frac{\delta(n)}{4} = \frac{\delta(n)}{4} \cdot 1 \geq \frac{\delta(n)}{4} \cdot \frac{\mathcal{A}(\Omega)^2}{4} = \frac{\delta(n)}{16} \mathcal{A}(\Omega)^2.$$

Multiplying by $\omega_n^{(n+2)/n}$ and using $E(\Omega) - E(B_1) = \omega_n^{(n+2)/n} D(\Omega)$ gives (44). $\square$

1.7 The near branch

Fix the dimensional working radius

$$R_*(n) := \max\{d(n), 2\}, \quad (45)$$

so $R_* \geq d(n)$ (the surrogates of Lemma 1.9 lie in B_{R_*} after translation, by Remark 1.10) and $R_* \geq 2$ (the hypothesis of Theorem 1.1).

Lemma 1.12 (Small-deficit transfer). *Set*

$$\tilde{c}_{\text{SV}}(n) := \omega_n^{-(n+2)/n} c_{\text{SV}}(n, R_*(n)) > 0. \quad (46)$$

For every open Ω with $|\Omega| = \omega_n$ and $D(\Omega) \leq \delta(n)$,

$$\mathcal{A}(\Omega)^2 \leq \left(\frac{4C_{\sharp}(n)}{\tilde{c}_{\text{SV}}(n)} + 4C_{\sharp}(n)^2 \right) D(\Omega). \quad (47)$$

Proof. Apply Lemma 1.9 to get $\tilde{\Omega}$ satisfying (21)–(24); by Remark 1.10 assume $\tilde{\Omega} \subset B_{d(n)} \subset B_{R_*(n)}$. Theorem 1.1 at $R = R_*(n)$ gives $E(\tilde{\Omega}) - E(B_1) \geq c_{\text{SV}}(n, R_*) \mathcal{A}(\tilde{\Omega})^2$; dividing by $\omega_n^{(n+2)/n}$ and using $|\tilde{\Omega}| = \omega_n$,

$$D(\tilde{\Omega}) \geq \tilde{c}_{\text{SV}}(n) \mathcal{A}(\tilde{\Omega})^2. \quad (48)$$

Combining (48) with (23),

$$\mathcal{A}(\tilde{\Omega})^2 \leq \frac{D(\tilde{\Omega})}{\tilde{c}_{\text{SV}}} \leq \frac{C_{\sharp}}{\tilde{c}_{\text{SV}}} D(\Omega).$$

Now use (24), the elementary $(a+b)^2 \leq 2a^2 + 2b^2$, and $D(\Omega) \leq \delta(n) \leq 1$ (so $D(\Omega)^2 \leq D(\Omega)$):

$$\begin{aligned} \mathcal{A}(\Omega)^2 &\leq (\mathcal{A}(\tilde{\Omega}) + C_{\sharp} D(\Omega))^2 \leq 2\mathcal{A}(\tilde{\Omega})^2 + 2C_{\sharp}^2 D(\Omega)^2 \\ &\leq \frac{2C_{\sharp}}{\tilde{c}_{\text{SV}}} D(\Omega) + 2C_{\sharp}^2 D(\Omega) \leq \left(\frac{4C_{\sharp}}{\tilde{c}_{\text{SV}}} + 4C_{\sharp}^2 \right) D(\Omega), \end{aligned}$$

which is (47). $\square$

Corollary 1.13 (Small-deficit quadratic bound). *With*

$$c_{\text{near}}(n) := \frac{\omega_n^{(n+2)/n}}{4C_{\sharp}(n)(1/\tilde{c}_{\text{SV}}(n) + C_{\sharp}(n))} > 0, \quad (49)$$

every open Ω with $|\Omega| = \omega_n$ and $D(\Omega) \leq \delta(n)$ satisfies $E(\Omega) - E(B_1) \geq c_{\text{near}}(n) \mathcal{A}(\Omega)^2$.

Proof. Rearrange (47) to $D(\Omega) \geq \mathcal{A}(\Omega)^2 / (4C_{\sharp}/\tilde{c}_{\text{SV}} + 4C_{\sharp}^2)$ and multiply by $\omega_n^{(n+2)/n}$, using $E(\Omega) - E(B_1) = \omega_n^{(n+2)/n} D(\Omega)$. $\square$

1.8 Proof of the global theorem

Proof of Theorem 1.2. Set

$$c_{\text{SV}}^{\text{glob}}(n) := \min \left\{ c_{\text{near}}(n), \omega_n^{(n+2)/n} \frac{\delta(n)}{16} \right\} > 0. \quad (50)$$

Let Ω be open with $|\Omega| = \omega_n$.

Case A: $D(\Omega) \geq \delta(n)/4$. Lemma 1.11 gives $E(\Omega) - E(B_1) \geq \omega_n^{(n+2)/n} (\delta(n)/16) \mathcal{A}(\Omega)^2 \geq c_{\text{SV}}^{\text{glob}}(n) \mathcal{A}(\Omega)^2$.

Case B: $D(\Omega) \leq \delta(n)$ (which includes $D(\Omega) \leq \delta(n)/4$). Corollary 1.13 gives $E(\Omega) - E(B_1) \geq c_{\text{near}}(n) \mathcal{A}(\Omega)^2 \geq c_{\text{SV}}^{\text{glob}}(n) \mathcal{A}(\Omega)^2$.

The two cases overlap on $\delta(n)/4 \leq D(\Omega) \leq \delta(n)$ and together cover $[0, \infty)$; in either case (2) holds with $c_{\text{SV}}^{\text{glob}}(n)$ as in (50), a dimensional constant because $\delta(n)$, $c_{\text{near}}(n)$ depend only on n (recall $R_*(n)$ and hence $c_{\text{SV}}(n, R_*(n))$ are dimensional).

General volume. For arbitrary finite-measure Ω put $\Omega_* := (\omega_n/|\Omega|)^{1/n} \Omega$, so $|\Omega_*| = \omega_n$, $\mathcal{A}(\Omega_*) = \mathcal{A}(\Omega)$ and $D(\Omega_*) = D(\Omega)$ by scale invariance of $\mathcal{A}$ and D . Applying (2) to Ω_* reads $D(\Omega) \geq \omega_n^{-(n+2)/n} c_{\text{SV}}^{\text{glob}}(n) \mathcal{A}(\Omega)^2$. Multiplying by $|\Omega|^{(n+2)/n}$ and using $E(\Omega) |\Omega|^{-(n+2)/n} - E(B^*) |B^*|^{-(n+2)/n} = D(\Omega)$ with $|B^*| = |\Omega|$, then $E = -T/2$, gives

$$T(B^*) - T(\Omega) = 2(E(\Omega) - E(B^*)) \geq 2c_{\text{SV}}^{\text{glob}}(n) \left(\frac{|\Omega|}{\omega_n} \right)^{(n+2)/n} \mathcal{A}(\Omega)^2,$$

which is (3). $\square$

Remark 1.14 (Sharpness of the exponent). The exponent 2 on $\mathcal{A}$ in (2) is optimal: it is saturated by smooth, volume-preserving ellipsoidal perturbations of B_1 , for which both $E(\Omega) - E(B_1)$ and $\mathcal{A}(\Omega)^2$ are of the same quadratic order in the perturbation parameter.

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